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AHE: Adaptive Homomorphic Encryption for Customizable Privacy in Heterogeneous Federated Learning

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***Abstract*—Homomorphic Encryption (HE) serves as a vital privacy-preserving technique in federated learning (FL) applications. However, while HE offers substantial privacy benefits, it also introduces significant computational and communication overhead. This challenge is further compounded in real-world heterogeneous scenarios, where clients vary in computing resources and privacy requirements. Existing solutions apply uniform static HE configuration to satisfy the privacy requirements of the most stringent client, resulting in high resource usage for all the clients. We propose AHE, a novel FL framework that integrates HE with adaptive privacy-aware clustering to dynamically optimize the latency-accuracy-privacy trade-off. AHE introduces two key innovations: (1) adaptive HE parameter management, which tailors encryption levels to individual client capabilities and customizable privacy needs, and (2) private cross-cluster aggregation, a lightweight key-switching based protocol that privately combines model updates across clusters with divergent HE schemes. To further minimize overhead, we design a parallelized key-switching mechanism that reduces server-side latency. Experimental results demonstrate that our approach enhances various FL applications by improving final accuracy up to 19% and reducing global round time up to 70%, significantly boosting training efficiency. Additionally, AHE is robust to dynamic client shift, and the parallelization strategy dramatically reduces server-side latency, enabling better scalability.**

***Index Terms*—Federated Learning, Homomorphic Encryption, Heterogeneous, Clustering.**

I. Introduction

Federated learning (FL) has emerged as a promising framework for privacy-preserving machine learning with model training across numerous clients without requiring direct data sharing [1]. However, recent studies have highlighted privacy concerns in FL [2], [3], particularly through attacks such as membership inference [4], which can lead to significant privacy leaks when model updates are shared [5]. To address these issues, homomorphic encryption (HE) [6], [7], differential privacy [8], multi-part computation [9] are utilized to encrypt local models, allowing for secure aggregation [10]. Among those approaches, HE enables direct operation (e.g., addition or multiplication) on encrypted data without decryption guaranteeing quantum security [11]. Although there are works on HE acceleration [12], how to adapt HE for FL in real-world heterogeneous setting still lacks study, where clients exhibit diverse resources and privacy demands.

HE introduces additional encryption noise during cryptographic operations (e.g., addition/multiplication), which accumulates with each cryptographic operation [13]. To control noise, HE uses parameters like *polynomial modulus* N and *coefficient modulus* q [14]. These parameters induce a trade-off: larger N and q enhance privacy (via stronger encryption) and precision but increase cryptographic overhead (encryption time) and communication costs. Within FL with HE, global model performance (or model accuracy) relies on precise HE encryption and operations. Existing FL systems enforce *uniform static* HE configurations applying large q and N-to satisfy the privacy and precision demand across all clients [15], but this approach ignores critical heterogeneity: 1) for resource-limited clients, large q and N cause higher encryption cost. 2) for privacy-sensitive clients, fixed settings prevent clients from gaining stronger privacy. To work for all clients, FL needs adaptive HE configurations: assign smaller HE parameters for resource-limited clients (reducing cryptographic latency), and larger HE parameters for privacy-critical clients (enhancing encryption strength).

When applying HE to real-world FL setting, the fundamental challenge lies in resolving HE's static settings with the dynamic and diverse requirements of heterogeneous FL systems. In real-world FL deployments, clients exhibit diverse resource capabilities (e.g., edge devices vs. cloud servers), varying privacy requirements (e.g., devices in government vs. customer), and non-independent and identical distributed (non-IID) data distribution—all of which demand dynamic non-uniform HE configurations. Yet, existing approaches enforce a uniform static HE setup, often selecting excessively conservative parameters to universally suppress HE noise while ensuring global model accuracy. This not only exacerbate straggler problem but also overlooks client dynamics.

Unfortunately, enabling customizable privacy with different HE parameters for each client independently is not possible, as ciphertexts generated under different parameter plans (e.g., varying N or q) become mutually incompatible, preventing cryptographic operations (i.e., aggregation on encrypted models) across clients. Thus, the core problem we address in this paper is twofold: (1) enabling fine-grained HE parameter adaptation to align with heterogeneous realities of FL, while (2) ensuring private collaboration across clients with divergent

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HE configurations.

Existing approaches to heterogeneous FL inadequately address the interplay between client heterogeneity and HE constraints: Clustering-based methods only consider client resources (e.g., CPU power) or data (e.g., gradient distribution) heterogeneity, ignoring how HE parameter adjustments influence their characteristics (e.g., latency, model accuracy). For example, to mitigate resources heterogeneity, FL+HC's latency-tiered clusters [16] only considering *round time* become obsolete, once HE parameters are optimized to reduce cryptographic overhead for resource-constrained clients. IFCA [17] and Auxo [18] cluster clients based on data distribution (e.g., gradient similarity), but overlook the resource heterogeneity. Client-selection based approaches [19], [20] priorize more efficient clients/clusters to mitigate resources and data heterogeneity simultaneously, but treat HE as a fixed, manually configured component. Critically, without decoupling the interplay between training resources, cryptographic capacity, and communication resources, HE optimization exacerbates disparities due to distinct influences across three dimensions [21]. Critically, HE optimization must address cryptographic and communication costs—two distinct dimensions directly governed by HE—while training latency, a core driver of client heterogeneity, operates independently of HE configurations; without a careful design of adaptive HE, none of them sufficiently address privacy heterogeneity (diverse privacy requirements) or support collaboration across clusters with divergent HE configurations.

TABLE I: Comparison of heterogeneity taming strategies.

Works	Resource			Data heterogeneity	Privacy heterogeneity
	train.	crypto.	comm.		
IFCA[17]	✗	✗	✗	✗	✗
FL+HC [16]	✓	✓	✓	✗	✗
Auxo [18]	✗	✗	✗	✓	✗
TiFL [19]	✓	✓	✓	✓	✗
Oort [20]	✓	✓	✓	✓	✗
AHE (Ours)	✓	✓	✓	✓	✓

Contribution. We propose AHE, the first FL framework to unify HE adaptation and client clustering in a single workflow, addressing the core challenge of incorporating cryptographic constraints with heterogeneous client needs. At its core, AHE introduces (1) privacy-aware clustering, grouping clients via a multi-dimensional metric (latency, privacy, accuracy needs) while incorporating the impact of HE; (2) a lightweight key-switching protocol enabling cross-cluster privacy-preserving aggregation of ciphertexts with diverse ciphertext; and (3) an adaptive trade-off mechanism that dynamically assigns cluster-specific HE parameters to optimize privacy-accuracy-latency trade-offs for individual requirements. Finally, self-correcting mechanism is included in AHE, which continuously refines clusters and HE settings, resolving misalignments caused by dynamic shifts. This end-to-end co-optimization eliminates manual tuning, sufficiently addressing all three kinds of heterogeneity (resources, data and privacy) in FL—key advances summarized in Table I. Experiments across image, language and speech demonstrate AHE 's superiority: AHE achieves up to 19% higher accuracy on non-IID data by aligning HE precision with cluster-specific needs, reduces up to 70% round time, and boosts training efficiency up to 87% over uniform static baseline. Under dynamic shifts (e.g., abrupt bandwidth constraints), its self-correcting mechanism recovers convergence 30% faster than static approaches, proving its efficacy in real-world heterogeneous FL environments.

Overall, we make the following contributions:

- **Privacy-aware clustering**. We introduce AHE tailored for heterogeneous FL environments, where it adapts clustering for HE-based FL. AHE integrates HE constraints and impacts into resource-based clustering, which conducts joint optimization of client groupings and HE configuration to improve converge efficiency.
- **Cross-cluster cryptographic consistency:** A cryptographic protocol enables privacy-preserving aggregation across clusters of disparate HE configurations without compromising efficiency.
- **Multidimensional trade-off automation:** Runtime adaptation of HE parameters to balance privacy, accuracy, and latency at both client and cluster levels.
- **Large scale real-world emulation:** Experimental evaluation based on real-world traces [22], [23] shows substantial performance improvements over existing solutions for enhancing the robustness and efficiency of FL systems.

Towards this end, we explain the following research questions throughout the paper.

RQ1 **Challenges.** Section III-A: How do HE-specific heterogeneity challenges (e.g., cryptographic incompatibility, dynamic client states) differ from conventional FL heterogeneity?

RQ2 **Clustering mechanism.** Section IV-B: How does privacy-aware clustering, which co-optimizes client groupings and HE parameters, overcome the limitations of prior FL clustering approaches?

RQ3 **Trade-off framework.** Section IV-C: What mechanism enables adaptive HE parameter selection to balance latency, accuracy, and privacy in heterogeneous setting?

RQ4 **System evaluation.** Section V-B: How does AHE perform in privacy-accuracy-latency trade-offs and convergence compared to baselines under customizable privacy requirements? How does AHE perform in terms of Section V-C: **overhead, scalability**?

II. Background and Related Work

This section discusses the relevant background in heterogeneous FL with HE that motivates our design for the studied threat model.

A. RNS-CKKS: Fully Homomorphic Encryption Scheme

Fully Homomorphic Encryption (FHE) enables both addition and multiplication on encrypted data, preserving privacy during outsourced processing. The Cheon-Kim-Kim-Song (CKKS) scheme [24] is an FHE method tailored for approximate arithmetic, making it ideal for privacy-preserving

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machine learning. Unlike exact FHE schemes, CKKS supports operation on fixed-precision real numbers, trading absolute accuracy for efficiency—a natural fit for numerical ML workloads. A key variant, RNS-CKKS, leverages the Residue Number System (RNS) to decompose large integer operations into parallelizable smaller modular computations, enhancing efficiency via SIMD-like batched processing [25]. This efficiency, coupled with CKKS's approximation-friendly design, has led to its adoption in encrypted ML inference and training [26]. In this work, "HE" refers specifically to the RNS-CKKS scheme.

Parameter Selection in RNS-CKKS. RNS-CKKS requires careful configuration of two critical parameters: 1) **Polynomial modulus** N: Determines the degree of the polynomial ring $\mathbb{Z}_N[X]/(X^N+1)$, directly impacting security and computational complexity. Larger N strengthens security but increases encryption time and memory usage. 2) **Coefficient modulus** q: A product of primes ($q = \prod q_i$) in RNS representation, governing the ciphertext modulus size. Larger q accommodates more noise growth during encryption and operations, improving numerical precision at the cost of cryptographic overhead (encryption cost) and privacy risk.

Encoding and noise dynamics. Real-number inputs are scaled by $\Delta \leq q$ to integers, quantized into plaintext polynomials, and encrypted via the Ring Learning with Errors (RLWE) problem [27]. Each operation introduces noise, which accumulates and reduces precision. To prevent noise overflow, q must suffice to restrict noise scale. However, overly large q risks privacy. For fixed N, excessively large q weakens privacy; thus, N and q must be jointly optimized under privacy guidelines like HE standards [14].

Balancing latency, accuracy, and privacy. In FL, model performance hinges on decryption precision, better precision preserves finer fractional details but demands greater q, increasing ciphertext size and encryption costs. Conversely, smaller q improves efficiency but risks accuracy loss. Furthermore, N and q are co-dependent: higher privacy levels require larger N, constraining the permissible q [28]. For example, with $N = 2^{13}$, q can safely span 218–438 bits, whereas $N = 2^{15}$ allows q up to 881 bits [14]. This interplay necessitates parameter tuning to align with accuracy, latency, and privacy requirements in deploying HE for FL.

B. Federated Learning

In FL, training data remains distributed across multiple devices. This approach significantly enhances data privacy and security, as sensitive information does not leave its original location [1], [29], [21]. During FL, the central server will select a subset of all clients as participants for the current round and send the current global model to them. Then, participants update models locally with their local data, and the updated local models are shared with the central server. The central server will aggregate the received models and update the global model. The procedures iteratively progress until convergence. However, in the real world, the resources are varying and limited, and non-IID data distributions make this convergence much more difficult to achieve [30].

Heterogeneous federated learning. In mitigating the challenges of heterogeneous FL, two main strategies emerge regarding system approaches. The first, client selection, is crucial for effective collaboration among diverse devices [21]. TiFL [19] enhances client selection by balancing accuracy with latency, selecting high-accuracy participants from latency clusters. Another approach, Oort [20], introduces a dynamic selection mechanism that prioritizes clients based on efficiency, thereby improving performance and accelerating convergence through an exploration-exploitation design. The other is cluster-based FL methods, which offer promising solutions to scalability and efficiency issues in FL [31]. CFL [32], FL+HC [16], and FedCHAR [33], focus on optimizing the aggregation process by leveraging hierarchical clustering techniques within FL frameworks. This innovative approach organizes clients into hierarchical clusters, thereby improving communication efficiency and convergence speed. Furthermore, TiFL [19] has a simple clustering mechanism that just splits the clients according to their latency order. Auxo [18] utilizes K-means to cluster the clients according to the uploaded gradients. However, none of them have addressed the heterogeneous challenges on the HE side.

III. Problem Overview

A. Challenges of Heterogeneous FL with HE

FL with HE provides a secure scheme for privacy-preserving collaborative model training, but faces significant challenges due to varying requirements and additional HE overhead [34], [35]. Key distribution in FL with HE involves distributing either the private key to clients, and the same public key and HE configurations to all participants before training begins. During training, participants encrypt their locally updated models using HE before sending them to the server, which averages these encrypted models. This process repeats until convergence [36].

To address **RQ1 Challenges**, the following challenges arise when considering clustering for heterogeneous FL with HE system, which even those advanced heterogeneity-aware approaches in FL is incapable to solve reasonably.

Interdependency between clustering and HE parameters. A fundamental challenge lies in the bidirectional relationship between client clustering and HE parameter optimization. Existing clustering methods, such as IFCA [17], FedAT [21] and FL+HC [16], group clients based on static characteristics like data distributions or resource profiles. However, in FL with HE, the assignment of HE parameters directly alters client characteristics. For instance, a resource-constrained "straggler" client may transit to a normal state after adopting lightweight HE parameters, rendering resource-only clustering assumptions obsolete. Clustering decisions must account for the effects of HE parameters, yet optimal HE parameters cannot be determined without prior knowledge of client groupings. Current approaches fail to address this feedback loop, as they assume client characteristics remain static post-clustering.

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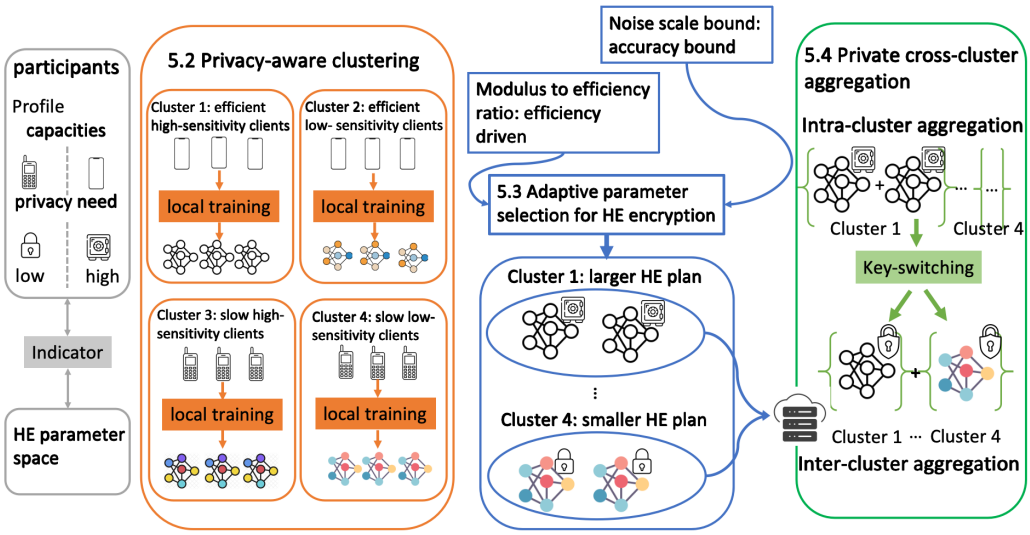


Fig. 1: Overview of AHE.

Needs for client-specific HE trade-offs. HE parameters non-linearly influence client latency, accuracy, and privacy, creating non-stationary system dynamics. For example, reducing the coefficient q to lower encryption cost for a slow client may degrade arithmetic precision, destabilizing model convergence. These effects introduce feedback loops between parameter selection, client performance, and system stability. Furthermore, conflicting optimization objectives—latency, accuracy, and privacy—require client-specific trade-offs. A uniform HE parameter plan risks over-provisioning privacy for some clients while under-protecting others.

Multi-objective optimization across clusters. Even with accurate clustering, divergent privacy and performance requirements across clusters complicate global optimization. Clusters prioritizing latency may adopt smaller N, while privacy-critical clusters may require larger N for stronger privacy. These incompatible HE configurations hinder HE cryptographic operations, as ciphertexts from different clusters cannot be directly aggregated.

B. Threat Model

In this part, we explain the threat model and formally define the privacy in HE. We consider scenario with a semi-honest adversary similar with [9], [15]. Semi-Honest Clients: A subset of clients may passively observe encrypted or aggregated model updates and seek to extract private data. Semi-Honest Server: The central server faithfully executes aggregation steps but aims to infer sensitive client information from intermediate computations, such as encrypted model updates. This framework assumes secure communication channels, excluding man-in-the-middle attacks.

Privacy Definition. The privacy guarantee of HE in FL is determined by its encryption strength (or security level), which corresponds to the computational hardness of solving the underlying RLWE problem [24] and is determined by the polynomial modulus N and coefficient modulus q. Following standardized HE security frameworks [11], we define privacy levels as *low* (128-bit security), *moderate* (192-bit), and *high* (256-bit), where higher privacy requires larger N and carefully bounded q. Under a semi-honest adversary $\mathcal{A}$ capable of corrupting clients or the server, HE ensures that corrupted clients learn no information about honest clients' encrypted data, while a corrupted server—without access to the secret key—cannot decrypt individual or global model updates.

IV. Adptive Homomorphic Encryption: Overview and Design

Building on the above insights, we introduce our adaptive HE approach for heterogeneous FL, named AHE, which integrates privacy-aware clustering, adaptive parameter optimization, and key-switching based aggregation. AHE adapts to data heterogeneity and enables better trade-offs in latency-accuracy-privacy.

A. Overview

To address the challenge of aligning heterogeneous client requirements with HE parameter configuration in FL, we propose a privacy-aware clustering framework that *jointly* optimizes client characteristics and cryptographic capability. Traditional approaches often overlook the gap between client resources and *uniform static* HE configurations, leading to inefficiency. As shown in Figure 1, AHE consists of three key components, privacy-aware clustering, adaptive parameter selection and private cross-cluster aggregation. Clustering is driven by two novel indicators that align client characteristics (efficiency and privacy demand) with HE parameter space, estimating the optimized HE configuration for each client under cryptographic constraints. This ensures intra-cluster consistency—where devices within the same cluster exhibit homogeneous needs and can adopt similar optimized HE parameters—while preserving inter-cluster adaptability to accommodate distinct cryptographic configurations across groups. To balance between latency, accuracy, and privacy, we design a dynamic trade-off mechanism that autonomously selects HE parameters at the cluster level, dynamically weighting

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cryptographic overhead against accuracy bounds and privacy guarantees. Crucially, a lightweight cross-cluster aggregation protocol bridges divergent HE configurations for aggregation.

B. Privacy-Aware Clustering

In this section, we outline our clustering method and motivate the design addressing **RQ2**. Our approach utilizes a one-step clustering, integrating latency, accuracy, and privacy into a two-dimensional map. We introduce two hyper-parameters to facilitate better trade-offs and parameter navigation: the accuracy penalty α, which help adapt HE to different training phase, and the privacy penalty δ, reflecting privacy needs.

Before clustering in each round, profiler iteratively collects most recent non-sensitive meta-data for client profiles, that is the total overhead T_i, the accuracy contribution P_i, privacy demand S_i (low, mid and high), and encryption cost L_i. To efficiently estimate improvement that local update would bring to the global model, we adopt a method similar to [20], using the size of training loss as a measure of potential accuracy contribution (although gradients directly capture the statistical heterogeneity, a larger gradient norm often attributes to bigger loss) without incurring additional computational costs. New available clients are initialized with a system preferred round time $T_i = T$, encryption cost $L_i = L$, and accuracy contribution $P_i = P$.

Privacy-aware indicators. To solve the clustering problem, we introduce a two-dimension indicator that integrates HE impact and client characters (latency, accuracy, and privacy) into a plane. The indicators are as follows.

$$
\begin{aligned}
I_{i,1} &= \underbrace{\frac{L}{\max\left(L_i,\, \psi(S_i)L\right)}}_{\text{Latency}} \cdot \underbrace{\left(\frac{P_i}{P}\right)^{\alpha \mathbb{1}(P_i<P)}}_{\text{Accuracy Penalty}}, \\
I_{i,2} &= \underbrace{\left(\frac{\phi(S_i)}{\phi_{\text{base}}}\right)^{\delta}}_{\text{Privacy}} + \underbrace{\lambda \cdot \frac{P_i}{L_i}}_{\text{Efficiency Tradeoff}},
\end{aligned}
\tag{1}
$$

where $\psi(S_i)$ scales latency via privacy (e.g., $\psi(\text{High}) = \phi_{base}/\phi(S_i)$), and $\phi_{base} = 128$ and $\phi(S_i)$ maps S_i to HE encryption strength (e.g., $\phi(\text{high}) = 256$). Hyperparameters α and δ control accuracy-privacy tradeoffs, λ is a constant aligning the range of efficiency with privacy, $\mathbb{1}(\cdot)$ is an indicator function (Note we focus on one round operation leaving out round index in the rest for simplicity).

The proposed scheme advances a co-design of client clusters and HE parameters, unifying system efficiency and cryptographic constraints into a single optimization manifold. Two complementary indicators, $I_{i,1}$ and $I_{i,2}$, which map client features to a latent space that align with HE parameter space. The interplay of these indicators ensures that clusters are *cryptographically compatible* (i.e., shared HE parameters satisfy privacy requirements) and *operationally coherent* (i.e., clients exhibit aligned latency-accuracy tradeoffs). At its core, this addresses the inter-correlation between HE parameter selection and client characteristics: the HE parameter N and q directly influence both cryptographic overhead L_i and accuracy contribution P_i, while client-specific privacy demands S_i impose upper bounds on q. By resolving these dependencies, the framework avoids the suboptimality of static approaches (e.g., resource-based clustering) and instead identifies configurations under cryptographic impact.

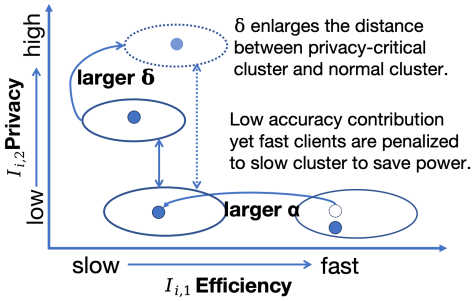


Fig. 2: Illustration of indicators. I_1 specifies potential range of ideal HE parameter, I_2 indicates the privacy needs.

The co-design's novelty lies in its explicit parameterization of the privacy-efficiency Pareto frontier. A pictorial of design intuition is shown in Figure 2. The term $I_{i,1}$ is dominated by a normalized clipped latency which aligns with value q. However, not all fast clients have a high accuracy contribution (e.g, clients with minimal dataset). To avoid those false-efficient clients occupy resources, we penalize their efficiency if their accuracy contribution is lower than preference P. Also, we use multiplication to scale latency part to preserve the range within (0,1), keeping alignment with parameter space. The term $I_{i,2}$ is dominated by $\left(\frac{\phi(S_i)}{\phi_{\text{base}}}\right)^{\delta}$ which imposes a strict partial ordering over privacy levels, ensuring clients with higher S_i occupy distinct regions of the latent space. However, the efficiency term $\frac{P_i}{L_i}$ introduces controlled privacy-efficiency trade-off: with $\delta = 1$ a client with moderate S_i but exceptional efficiency (e.g., $I_{i,2} = 192/128 + 1.5 \approx 3$) can attain an $I_{i,2}$ value comparable to a high-privacy client with baseline efficiency ($256/128 + 1 \approx 3$), effectively enabling vertical mobility across privacy tiers. This mechanism is emperically allow those more efficient clients to get the largest parameter space for N, q. The hyperparameter δ modulates this mobility, acting as a tunable coefficient between tiers: $\delta > 1$ enforces rigid stratification, while $\delta < 1$ permits fluidity, allowing the system to adaptively reweight privacy according to efficiency as federated objectives evolve.

Operationally, the scheme induces a 2-dimensional clustering topology. Clients are first partitioned along $I_{i,2}$ by their *privacy class* (determined by $\phi(S_i)$), then subdivided within each class by $I_{i,1}$ to group clients with homogeneous resource-cryptographic profiles. This hierarchy ensures cryptographic compatibility (clusters share HE parameters meeting all members' privacy requirements) while maximizing intra-cluster efficiency. The resultant clusters thus represent HE parameterized equivalence classes, where both the cryptographic configuration (N, q) and the operational performance (L_i, P_i) are jointly optimized.

Navigating the trade-off of latency and accuracy across the rounds. As the training progresses, the accuracy contribution will decrease, which makes all the clients have a

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relatively small $I_{i,1}$, leading to an inefficient training phase and degrading the final accuracy (due to insufficient participation/training). We enable a pacer $P \leftarrow P * r$ with $r < 1$ to rescale the preferred accuracy contribution, bargaining the efficiency and final accuracy. When participants that could reach the preferred accuracy contribution are less than a half, pacer reduces the preferred accuracy contribution. This approach avoids premature termination while ensuring low-accuracy clients are progressively incorporated in training. After that, clients are clustered to $|\mathcal{K}| = K$ groups $\mathcal{C}_\mathcal{K} = \{\mathcal{C}_k : k \in \mathcal{K}\}$ using online K-means [37] on their indicator vectors $(I_{i,1}, I_{i,2}, T_i)$, where T_i is the total overhead of client i. Note that to maintain consistent performance enhancement on both cryptographic and learning overhead, we separate clients' HE latency from their training time.

Clustering correction mechanism. To address adaption for dynamic FL setting, we choose to utilize online K-means. In unsupervised clustering, two prevalent challenges stem from the unknown prior data distribution: determining the optimal number of clusters and accurately clustering dynamic clients. To mitigate these issues, we include a system component specifically for clustering correction. This mechanism involves updating each client's profile at every iteration. Profile changes or unsatisfied privacy need in a client trigger a recalculation of the nearest cluster. Furthermore, if the clustering process exhibit slow convergence or a notable decrease in accuracy, we respond by re-clustering with an increasing number of clusters.

Unlike prior work that treats HE parameters as *uniform static* constraints, our method explicitly aligns the client capacity and HE optimization impact, enabling joint adaptation of clusters and HE configurations. This ensures clients are grouped not just by current characteristics, but by their latent compatibility under evolving HE parameter assignments.

C. Adaptive HE Parameter Selection for Clusters

In this section, we introduce a more refined HE parameter management approach to address **RQ3 Trade-off mechanism for parameter selection**.

We introduce an adaptive parameter assignment framework governed by two bounds: 1) Modulus-to-Efficiency Ratio (MER): $MER = \frac{Q(H)\cdot L}{P}$, where $Q(H)$ is the smallest modulus q that ensures the computation meets a target precision of H bits. In general, we choose H such that increasing it further would yield only marginal accuracy improvements. This ensures MER scales q to balance latency and accuracy without overprovisioning. 2) Noise-Sensitivity-Bound (NSB): NSB serves as a stricter lower bound to safeguard model efficacy (e.g., $NSB = Q(8)$ or FP8), which calibrates the maximum affordable noise.

For cluster $\mathcal{C}_k$, the q_k, N_k are computed as:

$$\begin{aligned} q_k &= \max(\underbrace{MER \cdot \bar{I}_k}_{\text{Latency}}, \underbrace{NSB}_{\text{Accuracy}}), \\ N_k &= \underbrace{N_{\min}(q_k, \bar{S}_k)}_{\text{Privacy}}, \end{aligned} \tag{2}$$

where $\bar{I}_k = \min\left(\frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k} I_{i,1}\cdot\frac{P}{L}, \frac{P}{L}\right)$ reflects cluster's efficiency and $\min(\cdot)$ avoids overflow, $\bar{S}_k$ is the highest privacy level within cluster k. $N_{min}(S, q)$ outputs minimum N that satisfies S for a given q. The $\max(\cdot)$ enforces $MER\cdot\bar{I}_k$ within the range $q_k \geq NSB$. Operationally, MER aligns efficiency $\bar{I}_k$ with the size of optimized q_k under least accuracy bound NSB, and N_k is determined by optimized q_k.

AHE introduces adaptive parameter stratification, where clusters autonomously negotiate q within privacy, accuracy-preserving bounds. Unlike uniform static approaches, this method dynamically reweights latency $\bar{I}_k$, accuracy floors NSB, and privacy limits S_i to maximize efficiency without compromising accuracy or privacy.

Algorithm 1 AHE: Adaptive HE Algorithm

Input: Index set of all clients $\mathcal{C}$, index set of all clusters $\mathcal{K}$ with size of K, accuracy penalty α, privacy penalty δ, accuracy pacer r. System preferred encryption cost L, preferred accuracy contribution P, encryption cost set L, accuracy contribution set P, privacy set S, and the train time set T, with subscription i indicates the correpsonding value of client i. The client indicator set I, modulus-to-efficiency ratio MER, Noise-sensitive-bound NSB

Output: Clustering setting: clients index $\mathcal{C}_k$, means m_k, HE parameter N_k, q_k for cluster k, and global model W_g

for each cluster $k \in \mathcal{K}$ **do**
  Generate switch key $(pk_{switch}, sk_{switch})$
  Initialize $\mathcal{C}_k^0 \leftarrow \emptyset,\ m_k^0 \leftarrow I_k^0$
for each round $t = 1, 2, \ldots$ **do**
  Sample clients $\mathcal{C}^t$ from $\mathcal{C}$
  for each client $i \in \mathcal{C}^t$ **do**
    $W_i^t \leftarrow HE.Dec(W_g^{t-1}, sk_{switch})$
    $W_i^t \leftarrow$ Client trains W_i^t with local dataset
    Profiler collects client i most recent characteristics $P_i^t, L_i^t, S_i^t, T_i^t$
    Server computes indictor $I_i^t = (I_{i,1}^t, I_{i,2}^t) \leftarrow$ INDICTOR$(L_i^t, P_i^t, S_i^t, P, L, \alpha, \delta))$ via equation (1). Add I_i^t to set I^t
  for each cluster $k \in \mathcal{K}$ **do**
    $\mathcal{C}_k^t \leftarrow$ SERVERCLUSTER$(C^t, \mathcal{C}_\mathcal{K}^{t-1}, I^t, T^t, P, L, \alpha, \delta)$
    $(N_k^t, q_k^t) \leftarrow$ PARAMSELECT$(\mathcal{C}_k^t, I^t, P, L, S^t,$ MER, NSB$)$
    Key center generate cluster key pair (pk_k^t, sk_k^t) and calculate KSK_k^t with (sk_k^t, sk_{switch})
  Each client $i \in \mathcal{C}^t$ sends $HE.Enc(W_i^t, pk_k^t)$ encrypting with assigned parameter (N_k^t, q_k^t) to server
  for each cluster $k \in \mathcal{K}$ **do**
    Server aggregates updates within cluster and switches aggregation to ciphertext under key sk_{switch}, $\widehat{W_k} = HE.Switch(\sum_{i\in\mathcal{C}_k^t} HE.Enc(W_i^t, pk_k^t), \text{KSK}_k^t)$ via equation (3)
  Server aggregates global model $HE.Enc(W_g^t, pk_{switch}) \leftarrow \sum_{k\in\mathcal{K}} \widehat{W_k}$
function SERVERCLUSTER$(\mathcal{C}^t, \mathcal{C}_k^{t-1}, I^t, T^t, P, L, \alpha, \delta)$
  if $|\{P_i^t > P, i \in \mathcal{C}^t\}| < |\mathcal{C}_k^{t-1}|/2$ **then**
    $P \leftarrow P \cdot r$
  for each client $i \in \mathcal{C}^t$ **do**
    if $\mathcal{C}_k^{t-1}$ is nearest to client i **then**
      $m_k^t \leftarrow m_k^{t-1} + \frac{1}{|\mathcal{C}_k^{t-1}|} * ((I_{i,1}^t, I_{i,2}^t, T_i^t) - m_k^{t-1})$
      Assign i to cluster $\mathcal{C}_k^t$
  return $\mathcal{C}_\mathcal{K}^t = \{\mathcal{C}_k^t\}_{k\in\mathcal{K}}$
function PARAMSELECT$(\mathcal{C}_k^t, I^t, P, L, S^t,MER, NSB)$
  $\bar{I}_k^t = \min\left(\sum_{i\in\mathcal{C}_k^t} \frac{I_{i,1}^t}{|\mathcal{C}_k^t|}\frac{P}{L}, \frac{P}{L}\right), \bar{S}_k^t = \max_{i\in\mathcal{C}_k^t}(S_i^t)$
  $N_k^t, q_k^t \leftarrow$ SELECT(MER, NSB, $\bar{I}_k^t, \bar{S}_k^t$) via equation (2)
  return (N_k^t, q_k^t)

D. Private Cross-Cluster Aggregation

To address the challenge of aggregating encrypted models across clusters with incompatible HE parameters, we propose

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a key- switching based aggregation protocol that enables private cross-cluster aggregation without decryption during aggregation. Each cluster operates with its own HE key pair (pk_k, sk_k), while a global *switching key pair* $(pk_{\text{switch}}, sk_{\text{switch}})$ configured with (N, Q) is pre-shared among all clients, where $Q = Q(10)$ and $N = 15$ similar to uniform static setting used to satisfy the highest privacy level. During aggregation:

- **Key-Switching Key Generation:** Each cluster computes over the keys sk_k and sk_{switch} to obtain the key-switching key KSK_k (according to section 2.2 in [38]). This key allows ciphertexts encrypted under sk_k to be converted into ciphertexts under sk_{switch} without decryption. After that, each client sends the KSK_k to server along with the local encrypted model $HE.Enc(W_i, pk_i)$.
- **Ciphertext Transformation:** After aggregating within cluster $\mathcal{C}_k$, the aggregation of $\mathcal{C}_k$ is transformed into ciphertexts compatible with sk_{switch} using the KSK_k. This ensures all gradients share a common encryption scheme without leaking information to server.

$$\begin{aligned}\widehat{W_k} &= HE.Enc(\sum_{i\in\mathcal{C}_k} W_i, pk_{switch}) \\ &= HE.Switch(Encrypt(\sum_{\mathcal{C}_k} W_i, pk_k), KSK_k)\end{aligned} \tag{3}$$

  where $Switch$ multiplies ciphertext by KSK_k.
- **Homomorphic Aggregation:** The server aggregates the ciphertexts $\widehat{W_k}$ under sk_{switch}, producing a globally encrypted model $HE.Enc(\sum_{k\in\mathcal{K}} W_k, pk_{switch})$.
- **Client-Side Decryption:** The aggregated result is broadcast to clients, where clients decrypt using their sk_{switch}.

This protocol ensures end-to-end encryption, allowing clusters to retain distinct HE configurations (e.g., varying ciphertext moduli q) while avoiding extensive cost of key-switching. The integration of key switching and HE-based clustering resolves the fundamental incompatibility between cryptographic constraints and client characteristic, ensuring latency, accuracy, and privacy coexist in heterogeneous FL systems. Note that, we don't use per-client key-switching as the time cost of key switching is relatively large slowing down central aggregation drastically, which is infeasible under cross-device setting. A comprehensive description of AHE is presented in Algorithm 1.

V. Evaluation

In this section, we conduct comprehensive experiments to address **RQ4 Evaluation** by answers the following key questions.

Q1. How much does the proposed AHE improve the training efficiency for the current FL selection algorithms?

Q2. How would the proposed clustering optimization method perform across the three aspects, i.e., latency, accuracy, and privacy?

Q3. Is the HE parameter selection design compatible with the clustering design, ensuring appropriate HE parameter choices for each cluster?

Q4. What is the impact of FL hyper-parameters on AHE's performance, and are there discernible guidelines for selecting them?

A. Experiment setup

Evaluation environment. In our investigation, we conduct large-scale, cross-device federated learning experiments utilizing NVIDIA RTX 4090 GPUs. To accurately simulate heterogeneous real-world scenarios, the dataset is partitioned using Dirichlet distribution [19]. Additionally, real resource traces from [39] are incorporated to simulate resources to each client. Our experimental framework is based on the open-source FL benchmark PFL [40] and HE tool [41], which uses a standard FL setup [22]. This setup ensures a realistic representation of both device capacities and data distributions. The efficacy of our federated learning system is quantified by measuring the wall clock time.

Datasets and models. We evaluate the effectiveness in cross-device scenarios through three popular applications on benchmark datasets: Image Classification with a ResNet-18 model [42] on Cifar-10 and CIFAR-100, Speech Recognition using the M5 model [43] on the AudioMNIST [44] and Google Speech Command dataset [45], and Language Modeling by fine-tuning DistilBert [46] on the SQuAD dataset [47] for question answering.

Implementation details. Our simulations involve extensive cross-device scenarios with 300 clients for the larger dataset Cifar-100, Google Speech and SQuAD, and 100 clients for the rest. Here, we use the traces from [22] to simulate the status of each device during the training, where idle time is added based on clients' resources. While each training round, only 10% of them will be selected by the FL system to participate in the FL training process. Efficient clients are prioritized to maintain the designated participant count. When each round is finished, we recycle participants for memory saving.

Parameters. We follow the standard parameter setting of the FL benchmark [40] for training and dataset partition. Specifically, we set the batch size to 10 and the initial learning rate to 2e-5 for the DistilBert model and 5e-3 for the others. We use regular system-recommend HE parameter $N = 2^{15}$ and $q = 420$ for baselines, and H is set to 10. We use the linear scaling rule to schedule the learning rate.

Baselines. We use the classical FL with *uniform static* HE configuration as a baseline named *baseline*. We also include a range of SOTA clustered FL approaches for comparison. resource-based clustering: FL+HC [16] and TiFL [19]; and data-distribution based clustering: IFCA[17] and Auxo [18], all of which are equipped with uniform static HE setup.

Metrics. Our evaluation metrics cover key aspects of the FL system, including final accuracy, efficiency, and cryptographic overhead. Specifically, speedup reflects the reduction in average round time compared to the *baseline*, while efficiency improvement quantifies the total time saved to reach the target accuracy. Accuracy refers to the global model's performance, such as with top-1 accuracy reported for image and speech classification tasks, and the F1 score for language modeling. Cryptographic overhead particularly refers to client side

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TABLE II: Summary of improvement over *baseline*. ('-' means the approach could not reach the target accuracy, where target accuracy is the highest accuracy achievable by all methods after full convergence excluding FL+HC.)

Task	Dataset	Target Acc.	Metric	FL+HC	TiFL	IFCA	Auxo	AHE
Images	Cifar10	44%	Speedup	56.2%	61.7%	15.3%	46.9%	**70.6%**
			Effi. Impr.	67.5%	60.0%	62.8%	73.9%	**87.5%**
			Final Acc.	44.7%	55.2%	55.1%	54.7%	**65.6%**
	Cifar100	32%	Speedup	13.8%	16.35%	15.8%	14.5%	**48.1%**
			Effi. Impr.	-	58.3%	25.8%	42.7%	**71.6%**
			Final Acc.	12.5%	32.2%	**33.0%**	32.6%	32.8%
Speech	AudioMINIST	77%	Speedup	43.3%	46.8%	26.6%	33.1%	**51.6%**
			Effi. Impr.	-	32.1%	35.3%	35.7%	**57.1%**
			Final Acc.	60.4%	95.9%	95.8%	96.1%	**96.4%**
	Google	70%	Speedup	43.0%	43.6%	33.4%	25.3%	**53.8%**
			Effi. Impr.	-	30.6%	49.8%	48.3%	**59.2%**
			Final Acc.	33.4%	73.6%	74.3%	75.2%	**76.8%**
Language	SQuAD	63%	Speedup	33.2%	36.8%	31.5%	34.2%	**48.6%**
			Effi. Impr.	53.2%	58.3%	54.7%	48.2%	**67.3%**
			Final Acc.	73.2%	73.9%	73.8%	73.3%	**75.3%**

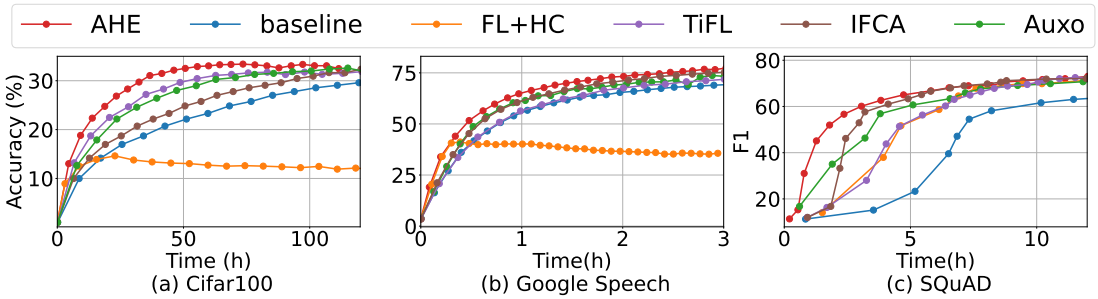


Fig. 3: Convergence lines of different clustering FL algorithms.

encryption and decryption overhead. Aggregation overhead refers to server side HE operations overhead.

B. Overall Performance

In this section, we address **Q5**, focusing on comparing AHE's performance against both the *baseline* (traditional FL with HE) and SOTA clustered FL solutions like FL+HC [16] and Auxo [18]. Furthermore, we fix the number of clusters at 20 and consistently use the TiFL's selector across all methods to maintain comparative fairness.

As shown in Table II, on CIFAR-10 (target: 45%), AHE reduces average round time by 70.6% (vs. $\leq$ 61.7% for baselines) and improves the early stage training efficiency by 87.5%, achieving 65.6% final accuracy. For CIFAR-100, AHE maintains robust performance with 58.9% speedup and 71.6% efficiency improvement, reaching 32.8% accuracy, while FL+HC collapses entirely (12.5% accuracy). In speech tasks, AHE achieves the highest efficiency (57.1% on AudioMNIST, 59.2% on Google Speech) and accuracy (96.4%, 76.8%), outperforming all other baselines. On SQuAD, AHE reduces round time by 48.6% and boosts training efficiency by 67.3% and achieves 75.3% accuracy, far exceeding the target.

Figure 3 (a)-(c) details the training process, illustrating significant improvements with AHE. The figure displays a sharper training trajectory for AHE, indicative of more efficient training and higher final accuracy. Notably, AHE outperforms FL+HC, which shows difficulty in converging, particularly in image and speech tasks. In the case of speech tasks, AHE's superiority is especially evident. Furthermore, all clustering approaches demonstrate substantial enhancements over the baselines in language model fine-tuning, highlighting the considerable potential of these methods.

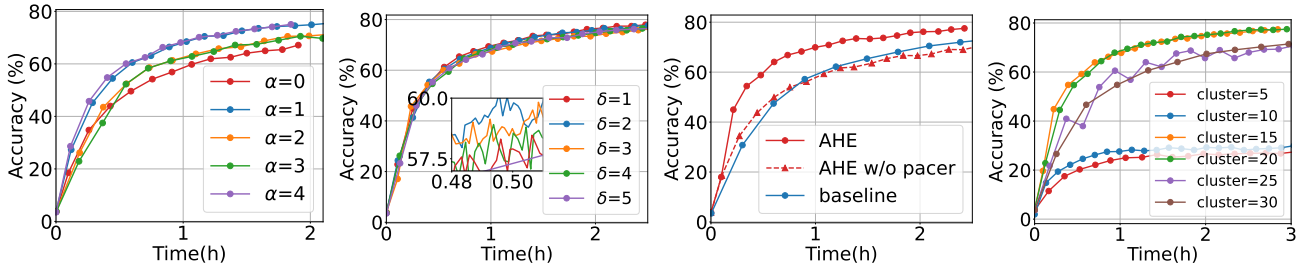
AHE consistently dominates in both per-round speed and end-to-end efficiency. For example, its 87.5% time savings on CIFAR-10 imply the target accuracy is reached in just 12.5% of the *baseline*'s total time, while baselines like IFCA and Auxo lag behind due to stragglers. Critically, AHE enables better trade-offs between speed and accuracy: it attains higher final accuracy while saving resource costs. Methods like FL+HC fail under complexity (e.g., 12.5% accuracy on CIFAR-100) or TiFL sacrifice efficiency for marginal gains (e.g., TiFL's 61.7% speedup vs. AHE 's 70.6%). These results underscore AHE 's unique ability to provide better trade-offs between low latency and high model accuracy.

C. Overhead Analysis.

In this section, we present a comprehensive breakdown of AHE's computational and communication overhead of both client and server components. We use the M5 trained on the Google Speech dataset as a case study for the rest part.

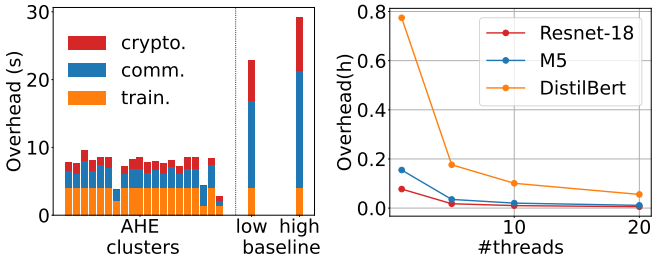
Client-side overhead of the AHE. In this experiment, we evaluated the latency of AHE relative to *baseline*, where all clients are configured with uniformed *low* or *high* HE plan. Figure 4(a) illustrates that AHE consistently shows lower latency across all clusters compared to the *baseline*, also resulting in reduced heterogeneity. This enhancement stems from AHE's HE parameter selection strategy, which balances efficiency without compromising privacy.

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(a) Impact of the accuracy penalty α. (b) Impact of privacy penalty δ. (c) Influence of Pacer. (d) Impact of #clusters.

Fig. 5: Sensitivity analysis.



(a) Client side overhead. (b) Server side overhead.

Fig. 4: Overhead analysis.

Parallelization of server-side computation. To further reduce the overhead of AHE, we integrate a parallelization mechanism into the server-side computation. Specifically, since ciphertext transformations are mutually independent across clusters, we assign each transformation to a dedicated thread for parallel execution. We evaluate the server-side overhead of AHE under varying thread counts to demonstrate its scalability (addressing **RQ4**). As shown in Figure 4(b), the server-side overhead decreases linearly as the number of threads increases. This design ensures efficient aggregation for large scale deployment.

D. Sensitivity analysis

In this section, we address **Q4** by conducting a comprehensive evaluation of the hyper-parameters within AHE.

Impact of accuracy penalty α. AHE employs the penalty factor α to prioritize high performance contribution participants by penalizing stragglers during selection. Figure 5(a) shows AHE's performance across different α values. While all nonzero α values outperform the no-penalty scenario ($\alpha = 0$), a larger penalty does not consistently enhance performance. This could be due to smaller HE parameters under higher penalties, leading to greater errors. Although the choice of α requires further refinement, its impact is still positive for the overall performance.

Impact of the privacy penalty δ. We explore the impact of the privacy penalty parameter (δ) on system efficiency and privacy. Figure 5(b) illustrates the trade-off between privacy and efficiency; however, the variations are minimal, potentially due to HE parameter optimizations that enhance system resilience. This trend suggests that AHE's efficiency is resilient to privacy heterogeneity. Despite subtle differences, the results affirm AHE's effectiveness in balancing strict privacy requirements, emphasizing its robustness.

Impact of the pacer. We investigate the impact of integrating a pacer into the clustering component of AHE, with findings illustrated in Figure 5(c). Initially, both AHE and its counterpart without the pacer (AHE w/o pacer) show quick improvements in model accuracy due to effective mitigation of stragglers and strategic client selection based on high performance contribution and system efficiency. However, as training progresses, AHE w/o pacer faces a slowdown in accuracy gains because it overlooks the diminishing performance contribution over time. In contrast, AHE with the pacer dynamically adjusts preferred performance contribution based on training progress, enhancing system effectiveness by balancing final accuracy against runtime constraints.

Impact of the number of clusters. In this experiment, we explore the influence of the number of clusters. Our results, as shown in Figure 5(d), demonstrate that AHE's performance varies across different cluster scales. Involving a too-large or too-small number of clusters leads to a degradation in both the efficiency and final accuracy. This can be attributed to the fact that (i) excessive number of clusters complicates the coordination and communication necessary for the training, leading to reduced efficiency (e.g., the number of clusters is 25 and 30 for their slow convergence). (ii) Inadequate clusters can not efficiently differentiate clients of different data distributions, incapable of tackling the Non-IID problem.

VI. Conclusion

In this paper, we present AHE a cluster-based HE approach for FL, while balancing the latency, accuracy, and privacy. Based on the designed indicators, AHE has adopted a compatible and adaptive strategy to manage the HE parameter, which helps the system make better trade-offs. A private yet efficient cross-cluster aggregation protocol is designed to enable combining the upload across divergent HE configuration. Compared to the existing works, AHE largely improves the efficiency and adapts to diversion applications.

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